

VOLUME I - PROBLEM RESEARCH / DOCUMENT 2 OF 3

Existing Challenges

Failure Modes of Current Controls and Constraints on Any Deployed Solution

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ABSTRACT

Volume I Document 1 established why fog constrains Bailadila production. This document examines why the controls presently in use do not resolve it. Each existing control is assessed against the specific physics and operating conditions of a dense-fog hilltop haul road, and the residual risk is identified. The second half of the document catalogues the environmental, infrastructural, human and organisational constraints that any proposed system must survive - these are the conditions that have historically caused promising mining technology pilots to fail at the transition to production use.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. Assessment Framework

Each control in current use is assessed on five axes. The assessment is deliberately unsentimental: several of these controls are mandatory, well-engineered and genuinely effective in the conditions for which they were designed. The question asked here is narrower - what does each control deliver at a meteorological optical range of 3-5 m on a graded, curved, two-way haul road?

Axis	Question asked
Physical validity	Does the underlying physics still function at extinction coefficients of 600-1000 per km?
Coverage	Which hazard classes and which geometries does it address?
Latency	How quickly does it convert a hazard into an operator-actionable signal?
Residual risk	What remains uncontrolled after it is correctly applied?
Production impact	Does it permit continued operation, or does it only make stopping safer?

The final axis is the one most often omitted from safety reviews and is central to this programme. A control that improves safety by making haulage stop is a legitimate control, but it is not a solution to the problem as stated.

2. Current Engineering Controls and Their Limits

2.1 Vehicle lighting and fog lamps

What it does. Headlamps, auxiliary fog lamps and amber beacons increase the luminous flux emitted into the forward scene, and auto-dipping systems reduce glare to oncoming operators.

Why it fails in dense fog. Lighting is subject to a physical trap that is counter-intuitive and frequently misunderstood by operators. The emitted light must traverse the fog twice - outbound to the target and back to the eye - so the received signal falls as the square of the one-way transmittance. Meanwhile, the droplets immediately in front of the lamp backscatter a fraction of the outbound beam directly into the operator's field of view. This backscatter is unattenuated by distance and forms a bright veil that raises the background luminance against which the target must be discriminated.

The result is that **increasing lamp intensity in dense fog reduces contrast rather than improving it**. Operators observe this directly: switching to high beam in thick fog makes the situation worse. Amber fog lamps mounted low and wide mitigate the geometry somewhat by reducing the overlap between the backscatter cone and the line of sight, but they do not change the underlying relationship. At MOR 3-5 m the achievable improvement is a small number of metres.

Residual risk. Effectively total for forward hazard detection at any useful haul speed.

Production impact. Nil. Lighting does not enable continued operation in dense fog.

2.2 Reflective delineation and road marking

What it does. Retroreflective posts, edge delineators, painted or banded lane markers and reflective berm markers give the operator a lateral reference for the running surface.

Why it fails. Retroreflective delineation suffers the same double-traverse attenuation as lighting, plus an additional penalty: the reflector's own return is a small fraction of the incident flux. In practice, delineators become visible only when they are already nearly abreast of the cab. They provide *confirmation* of position, not *anticipation* of the road ahead. At MOR 3-5 m, a driver navigating by delineators is effectively feeling along the road edge.

A second, more serious problem is site-specific. Bailadila haul road surfaces carry heavy iron ore fines. During the monsoon these fines become a wet, adhesive red slurry that coats every vertical surface. **Retroreflective devices on an iron ore haul road lose a large fraction of their performance within days of installation** and require a cleaning regime that is itself difficult to sustain in the monsoon. Field experience on ore haul roads generally is that maintained reflectivity is the exception rather than the rule.

Residual risk. High for lateral position; total for forward hazards. Delineation says nothing about what is on the road.

Production impact. Marginally positive - it permits crawling movement where none would otherwise be possible.

2.3 Rear-view cameras and mirrors

What it does. DGMS Technical Circular 03 of 2024 requires rear-view cameras on all opencast mine dumpers. They address the well-documented rear blind zone of a large rear-dump truck.

Why it fails in fog. A rear-view camera is a visible-band optical sensor. It is subject to exactly the same Mie scattering limit as the human eye, with the additional handicaps of a smaller aperture, limited dynamic range against a bright fog veil, and a lens that accumulates ore-laden water spray. Mirrors fare worse still, since they are unheated, unwiped and continuously fouled.

Residual risk. The rear blind zone, which the control was specifically installed to address, is uncontrolled in exactly the conditions where the risk of a person or light vehicle being in it is greatest - because in fog, that person or vehicle also cannot see the truck.

Production impact. Nil in fog.

2.4 Reverse alarms and audio-visual warning

What it does. Audio-visual alarms per DGMS Circular 08/2017 and Regulations 107/108 of MMR provide a distinct audible signal on start-up and reversing, supported by flashing beacons.

Why it is only a partial control. Audio propagation is essentially unaffected by fog, which makes this the one legacy control whose physics survives the conditions. Its limitations are different in nature:

- ? **Omnidirectional and non-informative.** An alarm tells a listener that a machine is reversing somewhere. It conveys no range, no bearing and no identity. In a congested loading area with multiple machines, the listener cannot determine which machine is the threat.
- ? **Masked by the acoustic environment.** Haul road ambient noise from engines, the truck's own powertrain, heavy rain on a steel canopy and the cab's own sealing all compete with the alarm. An operator inside a closed, air-conditioned cab may not hear an external alarm at all.
- ? **Habituation.** In a busy pit, reverse alarms sound continuously. Habituation is well documented and reduces the alerting value of the signal over a shift.
- ? **Beacons are optical.** The visual half of the audio-visual pair is attenuated identically to all other lighting.

Residual risk. Moderate to high. The alarm protects alert pedestrians at short range in quiet conditions. It does not protect a truck operator from another truck.

Production impact. Nil.

2.5 Existing proximity warning devices

What it does. DGMS Technical Circular 06 of 2020 requires proximity warning devices that detect static and moving objects around a dumper, provide audio-visual warnings, define a virtual target area, and scale the warning with speed and direction. Commercial implementations commonly use short-range ultrasonic sensors,

low-cost radar, RFID or UWB tags, or camera-based AI pedestrian detection.

Why it is insufficient for this problem. Existing proximity systems as commonly deployed are designed for a different hazard: the **static or low-speed blind-zone interaction**, typically within 10-15 m of the machine, chiefly protecting pedestrians during manoeuvring and reversing.

The dense-fog haul road problem is a **dynamic, high-closing-rate, long-range** problem. Consider a loaded truck descending a ramp. To stop safely it needs warning at a distance that accounts for perception-reaction time plus the braking distance of a heavy vehicle on a wet gradient - a distance measured in tens of metres, not in single-digit metres. A proximity device with a 10 m envelope issues its warning after the collision is already unavoidable.

Additional limitations of the deployed generation:

- ? **Tag-dependent systems** (RFID, UWB, magnetic-field personal tags) only detect what carries a tag. A rockfall, a stalled contractor vehicle, an unregistered light vehicle, a washed-out road edge or a buffalo on the road is invisible to them. In monsoon conditions, non-cooperative hazards increase rather than decrease.
- ? **Camera-based AI detection** is a visible-band optical modality and fails with the camera.
- ? **Ultrasonic sensors** have very short range, are degraded by wind and heavy rain, and are unsuited to haul speeds.
- ? **No shared picture.** A proximity device warns one operator. It does not tell the approaching vehicle, and it does not tell the control room.
- ? **Alarm fatigue.** Systems with poorly tuned envelopes generate frequent nuisance alarms in congested areas, and operators learn to disregard them - a documented cause of proximity system failure in practice.

Residual risk. High for the specific haul road collision archetypes identified in Volume I Document 1, Section 5.

Production impact. Low. A short-range warning device does not give an operator the confidence to maintain speed.

2.6 GPS fleet management and dispatch

What it does. Fleet management systems provide vehicle location, dispatch assignment, cycle time recording, payload and fuel data, and productivity reporting. Positioning is typically standard GNSS.

Why it does not solve the problem. Fleet management and collision avoidance have different accuracy and latency requirements, and the gap between them is decisive.

Requirement	Typical FMS capability	Requirement for fog guidance
Positional accuracy	2.5 to 10 m CEP (standard GNSS)	Sub-metre, ideally 10-30 cm, for lateral lane position
Update rate	1 Hz to once per 30 s, often lower to save bandwidth	5-20 Hz for closing-rate computation
Latency to other vehicles	Via server round trip; seconds	Direct vehicle-to-vehicle; tens of milliseconds
Availability	Degrades in pit shadowing; gaps tolerated	Must be continuous and integrity-monitored
Non-cooperative hazards	Not sensed at all	Must be sensed

A 5 m positional uncertainty is larger than the lane width available to a haul truck. **Standard GNSS cannot tell an operator which side of the road they are on.** Additionally, deep pit geometry and the Bailadila ridge produce satellite shadowing and severe multipath from high rock faces, degrading the already insufficient accuracy exactly where the road is most constrained.

Residual risk. Complete for collision avoidance; FMS was never intended to provide it.

Production impact. Positive but indirect. FMS gives the control room visibility of the productivity loss; it does not reduce it.

2.7 Fixed CCTV on haul roads

What it does. Pole-mounted cameras at junctions, tipping points and critical curves give the control room a view of traffic.

Why it fails. Optical modality; fails with visibility. Fixed installations on an ore haul road also suffer rapid lens fouling from ore dust in the dry season and mud spray in the wet season, and require power and backhaul to remote ridge positions. In dense fog a CCTV wall shows uniformly grey frames - which is operationally worse than no camera, because it creates the impression of monitoring where none exists.

Residual risk. Total in the target conditions.

2.8 Radio voice communication

What it does. VHF/UHF voice radio allows operators to report position, request passage on single-lane sections and coordinate with the control room. Radio propagation is unaffected by fog.

Why it is insufficient. Voice coordination is the traditional fallback and it does work - at very low traffic density. Its limits are structural:

- ? **Serial channel.** One transmission at a time. As vehicle count rises, the channel saturates. Calling positions for a fleet of dozens of machines is not feasible.
- ? **Human latency.** Perceive, formulate, key, transmit, receive, interpret, act. Several seconds at best. Closing rates on a haul road do not permit it.
- ? **Position by verbal landmark.** "I'm near the second bend above the ramp" has an uncertainty of tens of metres and depends on shared mental maps that new or contract operators may not have.
- ? **No non-cooperative hazards.** A washout or a rockfall is reported only if someone has seen it - which requires visibility.
- ? **Language and workforce mix.** Contract operator populations with varied first languages add error under stress.

Residual risk. Moderate at low density, high at production density.

Production impact. This is the control that currently permits limited movement in fog, at heavily reduced speed. It is the incumbent that any new system must beat.

3. Current Administrative Controls and Their Limits

Administrative controls in use typically include reduced speed limits during declared adverse weather, convoy or pilot-vehicle movement, one-way traffic rules on critical sections, spotters and signallers at junctions and reversing areas, suspension of light vehicle movement, and full suspension of haulage below a visibility threshold.

These controls are correct and should be retained. Their limitations for this programme are:

- ? **Binary and conservative.** A visibility threshold assessed by human judgement produces a go/no-go decision. Because the assessment is subjective and the consequence of being wrong is severe, the threshold is applied conservatively - correctly so, but at a production cost.
- ? **Spatially uniform, when fog is not.** A declaration applies to the mine or a road. Fog at Bailadila is strongly stratified by elevation and patchy in the advection regime. A hilltop bench may be in dense cloud while the

lower ramp is clear. A single declaration either stops clear sections unnecessarily or permits movement in a section that is not clear.

- ? **Spotters are exposed.** Placing a person on foot on a haul road in dense fog to guide vehicles transfers risk to the most vulnerable participant. This is the least defensible of the current controls.
- ? **Pilot vehicles constrain the whole convoy.** A light vehicle leading a haul truck sets the speed of the entire string, and is itself a collision target.
- ? **Restart overhead.** As set out in Volume I Document 1, repeated declare/stop/restart cycles impose fixed costs that can exceed the stoppage durations.

The central finding of Section 2 and Section 3 together: **every engineering control currently deployed on the haul road either operates in the visible band, or operates at a range far shorter than the stopping distance of a loaded truck, or provides only a single-operator warning with no shared picture.** The single control whose physics survives - voice radio - is limited by human latency and channel capacity. This is the gap the proposed solution must fill.

4. Environmental and Physical Constraints

Any hardware placed on a Bailadila haul truck or haul road must survive the following. These are not incidental deployment details; several of them have historically been the proximate cause of mining technology pilots failing.

Constraint	Specification implication
Ore dust, dry season	IP66 minimum, preferably IP67/IP69K on external units; conformal-coated PCBs; sealed connectors with retention
Monsoon water ingress and driven rain	IP67/IP69K; drainage paths; no upward-facing unsealed apertures
Adhesive iron ore fines slurry	Self-shedding or wiper-equipped sensor apertures; radar preferred over optical because radar tolerates a fouled radome far better than a lens tolerates a fouled window
Vibration and shock on HEMM	Compliance with vehicular shock and vibration standards; no hard-disk storage; strain-relieved harnessing; anti-vibration mounts
Temperature range	-5 deg C to +65 deg C ambient; cab and engine-bay mounted electronics exposed to higher soak temperatures
Humidity	Sustained 95% RH with condensation cycling; conformal coating and internal desiccation
Electrical environment	24 V nominal, wide-range DC input, load-dump and transient protection to vehicular standards; galvanic isolation from machine CAN
Lightning and surge	Exposed ridge infrastructure requires surge protection on all outdoor cabling and antenna feeds
Ferrous environment	Massive conductive masses (truck bodies, ore stockpiles, high-grade haematite faces) distort RF propagation, cause multipath and affect magnetic sensing
Blasting	Periodic overpressure and ground vibration; equipment and roadside infrastructure must be protected or relocatable
EMC	Coexistence with existing radio, blasting initiation systems and plant electrical equipment; no interference with statutory systems

The ferrous environment constraint deserves emphasis because it is specific to high-grade iron ore and is often overlooked when adapting automotive or generic mining technology. Bailadila haematite is among the highest-grade in the world. Large masses of conductive ore produce strong RF reflections and multipath, degrade GNSS through signal reflection off high faces, and can interfere with magnetometer-based heading references. Any RF or magnetic subsystem must be validated *on site*, not in a generic mining environment.

5. Infrastructure Constraints

- ? **Connectivity.** Bailadila is remote and hilly. Cellular coverage is limited and non-uniform. Ridge-and-pit geometry produces deep RF shadows. Any solution that assumes continuous broadband connectivity to every vehicle will fail. Design must tolerate intermittent backhaul, with vehicle-side autonomy and store-and-forward behaviour.
- ? **Power at roadside positions.** Mains power along haul roads is limited. Roadside infrastructure should be solar-plus-battery capable, which in turn constrains the power budget of any roadside sensing or communication node - a material constraint during a monsoon with persistent heavy cloud, when solar yield is at its annual minimum precisely when the system is most needed. Battery autonomy must be sized for multi-day low-insolation periods.
- ? **Haul road mutability.** Haul roads in an active open cast mine are not fixed civil assets. Ramps are extended, benches advance, routes are re-cut. **Any solution requiring fixed roadside infrastructure at close spacing incurs a recurring relocation cost** and risks being a permanent obstruction to mine development. This argues strongly for vehicle-borne sensing with minimal, strategically placed roadside assets.
- ? **Survey control.** Delivering sub-metre positioning requires a local reference network - DGNSS/RTK base stations or a CORS link - with adequate baseline coverage over the ridge and pit, and a defined datum tied to the mine survey grid.
- ? **Existing systems.** NMDC operates established fleet management, weighbridge, dispatch and plant control systems. A new system that does not integrate with them creates a parallel data estate and a second set of screens in the control room, which is a known driver of non-adoption.

6. Human Factors Constraints

This is the constraint class most often underestimated, and the one most likely to determine whether a technically sound system delivers value.

- ? **Operator workload.** A haul truck operator on a wet descent in fog is at high cognitive load. Any interface that demands reading, interpretation or interaction adds to that load. Information must be conveyed pre-attentively - by spatial position, colour, motion and sound - not by text.
- ? **Glance time.** Automotive human-factors practice constrains single glances away from the forward scene to roughly two seconds. The design target should be shorter. In dense fog the operator's visual attention is straining against the windscreen; a display must reward a very brief glance.
- ? **Trust calibration.** Two opposite failures are possible. **Under-trust** leads to operators disregarding the system and reverting to crawling, eliminating the production benefit. **Over-trust** leads to operators driving faster than the system's true capability supports, creating a new hazard that did not previously exist. Over-trust is the more dangerous and is a documented failure mode of driver assistance systems generally.
- ? **Alarm fatigue.** The dominant cause of proximity system rejection in mining. A system that alarms frequently in normal congested operation will be ignored or disabled. Alarm logic must be graded, contextual and speed-and-trajectory aware, not a simple distance envelope.

- ? **Workforce mix and literacy.** A substantial contractor operator population with varied language backgrounds and varied technology familiarity. Interfaces must be predominantly non-textual and trainable in a short induction.
- ? **Cab environment.** High noise, vibration, gloves, and a display that must be readable both in bright glare and at night without destroying dark adaptation.
- ? **Industrial relations.** Any system capable of monitoring individual operator behaviour raises legitimate concerns about surveillance and performance management. Experience across the sector is consistent: systems introduced as safety tools with transparent, agreed data-use policies are accepted; systems perceived as disciplinary instruments are resisted, disabled or defeated. Data governance is a deployment-critical design issue, not an afterthought.

7. Organisational and Commercial Constraints

- ? **Retrofit, not new build.** The solution must fit an existing mixed fleet of varying makes, ages and configurations, including machines whose OEM does not offer an integration path.
- ? **Fleet heterogeneity.** Different truck classes, dozers, graders, water tankers, light vehicles and personnel carriers. A collision avoidance system is only as good as its coverage; partial fleet fitment leaves uninstrumented machines invisible to the system, which can be actively dangerous if operators come to rely on the display.
- ? **Installation window.** Machines are fitted during planned maintenance. Installation must be achievable in a short, defined window with a documented harness and no machine-specific engineering.
- ? **Maintainability.** Field-replaceable units, self-diagnostics, and fault reporting to the control room. Any system that requires a specialist to diagnose will accumulate silent failures. A silently failed collision avoidance system is worse than none.
- ? **Statutory approval.** Advisory systems sit comfortably within current DGMS practice. Any function that takes control authority - automatic braking, enforced speed limiting, autonomous operation - requires a defined approval, testing and permissioning pathway with DGMS, with lead time that must be built into the programme plan.
- ? **Security context.** Bailadila operates in a security-sensitive region. This constrains deployment logistics, contractor access and the siting of remote unattended infrastructure, and raises the value of tamper-evident and self-reporting equipment.
- ? **Total cost of ownership.** Per-vehicle cost multiplied by a large mixed fleet, plus infrastructure, integration, training and a recurring maintenance and calibration burden. Cost per vehicle is a first-order design constraint, which biases the solution toward automotive-grade components at automotive volumes rather than bespoke mining hardware.

8. Why Prior Pilots Fail - Failure Mode Catalogue

Drawn from the general record of technology introduction in mining, the following failure modes should be treated as active risks to be designed against.

#	Failure mode	Mechanism	Design counter-measure
F1	Alarm fatigue	Envelope-based alarms fire constantly in congested areas; operators disregard or disable	Trajectory- and speed-aware graded alerting; measure and tune false alert rate as a headline KPI
F2	Sensor fouling	Optical apertures obscured by ore mud within hours; system degrades silently	Radar-first architecture; fouling self-detection; declared degraded mode

#	Failure mode	Mechanism	Design counter-measure
F3	Silent failure	A unit fails; nobody knows; operators still trust the display	Continuous self-test, heartbeat to control room, mandatory pre-shift verification integrated with DGMS checklists
F4	Partial fitment	Uninstrumented machines invisible; operators assume the display is complete	Non-cooperative sensing as a mandatory layer; explicit interface indication of cooperative vs sensed contacts
F5	Over-trust	Operators drive faster than system capability supports	Display sensing range and confidence explicitly; couple advisory speed guidance to measured visibility and system health
F6	Connectivity assumption	Design assumes continuous link; RF shadows break it	Vehicle-side autonomy; direct V2V independent of infrastructure; store-and-forward
F7	Integration island	Parallel system, second screen, duplicate data; control room ignores it	Integrate with existing FMS and dispatch; single operating picture
F8	Relocation burden	Dense roadside infrastructure obstructs mine development and is costly to move	Minimise fixed assets; make any roadside node portable and self-powered
F9	Industrial relations	Perceived as surveillance; resisted or defeated	Transparent, agreed data governance from the outset; introduce as a safety tool
F10	Calibration drift	Multi-sensor extrinsics drift with vibration; fusion quality degrades unnoticed	Online extrinsic calibration monitoring; automated alert on drift
F11	Validation gap	System tested in clear weather, deployed for fog	Mandatory validated fog-condition acceptance testing with measured MOR
F12	Pilot-to-scale collapse	Works on five trucks with vendor engineers present; fails on 200	Design for scale from the outset; automated provisioning, remote diagnostics, documented field maintenance

F11 deserves particular attention. It is trivially easy to demonstrate a radar-based collision avoidance system on a clear day and conclude that it works. The entire value of this programme rests on performance in conditions that occur during a limited annual window. Acceptance testing must be scheduled against the monsoon calendar, with instrumented visibility measurement, or the programme will be accepted on evidence that does not address its purpose.

9. Consolidated Constraint Register

The following register consolidates the constraints that Volume I Document 3 converts into formal requirements.

Ref	Constraint	Class	Severity
C-01	Visible-band sensing is non-functional at MOR 3-5 m	Physical	Critical
C-02	NIR LiDAR range collapses in dense fog	Physical	Critical
C-03	Required warning range exceeds existing proximity device envelopes	Physical	Critical
C-04	Standard GNSS accuracy is coarser than available lane width	Physical	Critical
C-05	Pit and ridge geometry cause GNSS shadowing and multipath	Physical	High
C-06	High-grade ore mass distorts RF propagation and magnetic heading	Physical	High

Ref	Constraint	Class	Severity
C-07	Ore-laden mud fouls optical apertures within hours	Environmental	High
C-08	Sustained high humidity with condensation cycling	Environmental	High
C-09	HEMM shock and vibration environment	Environmental	High
C-10	Lightning and surge exposure on ridge infrastructure	Environmental	Medium
C-11	Non-uniform cellular and RF coverage	Infrastructure	High
C-12	Limited roadside power; minimum solar yield during monsoon	Infrastructure	High
C-13	Haul road geometry changes with mine development	Infrastructure	High
C-14	Must integrate with existing FMS, dispatch and plant systems	Infrastructure	Medium
C-15	Operator cognitive load already high; glance time under 2 s	Human	Critical
C-16	Alarm fatigue risk	Human	Critical
C-17	Trust calibration - under-trust and over-trust	Human	High
C-18	Mixed-language, mixed-familiarity contract workforce	Human	High
C-19	Data governance and surveillance concerns	Organisational	High
C-20	Retrofit across heterogeneous mixed fleet	Organisational	Critical
C-21	Short installation windows within planned maintenance	Organisational	Medium
C-22	DGMS approval pathway required for any control authority	Regulatory	Critical
C-23	Compliance with MMR 1961 and DGMS Circulars 06/2020, 03/2024	Regulatory	Critical
C-24	Security-sensitive region constrains logistics and remote assets	Organisational	Medium
C-25	Cost per vehicle across a large fleet	Commercial	High
C-26	Acceptance must be validated in measured fog conditions	Programme	Critical

Volume I Document 3 translates this register, together with the problem statement of Document 1, into a structured field requirements specification.